Chapter 1 : Reaction Kinetics

PSPM 2011 / 2012

Q. 1 (a) A radioactive element ***K*** with an initial concentration of 3.5×10^{-6} mol L^{-1} undergoes decomposition reaction at 25°C. The rate constant of the reaction is 3.17 $year^{-1}$.

Determine the order of decomposition reaction of ***K***. Explain your answer.
Calculate the concentration of element ***K*** after 6 months and the time taken for the concentration to reduce to 1.75×10^{-6} mol L^{-1}. 8m

Answer (a)

i) order of reaction is first order
- rate constant, *k* unit is $year^{-1}$ which is characteristic of first order reaction

ii) $\ln \left(\frac{[K]_o}{[K]_t}\right) = kt$

$$\ln\left(\frac{3.5 \times 10^{-6}}{[K]_t}\right) = \frac{(3.17)}{1\ \text{year}} \times 6\ \text{month} \times \frac{1\ \text{year}}{12\ \text{months}}$$

$$[K]_t = 7.173 \times 10^{-7}\ \text{mol L}^{-1} / \text{M}$$

iii) since the final concentration is halved of the original concentration then $t = t_{½}$

$$t_{½} = \frac{\ln 2}{K} = \frac{\ln 2}{3.17\ \text{year}^{-1}} = 0.22\ \text{year}$$

Or

$$\ln \frac{[A]_o}{[A]_t} = kt$$

$$\ln \frac{(3.5 \times 10^{-6})}{(1.75 \times 10^{-6})} = kt$$

$$\ln 2 = kt$$

$$t = \frac{\ln 2}{3.17} = 0.22\ \text{year}$$

Q. 1 (b) Define activation energy.

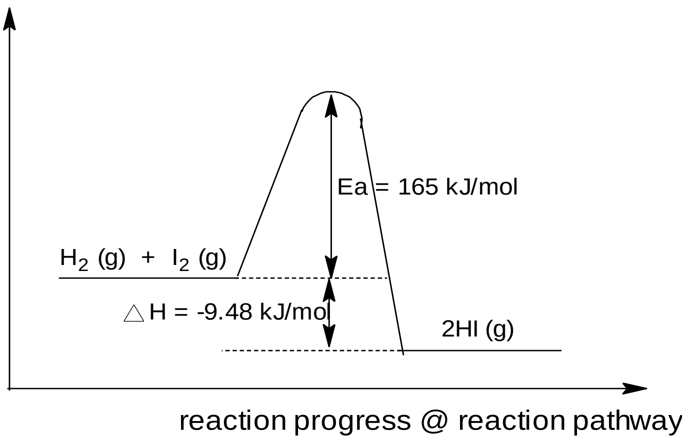
The rate constant for the forward reaction of a reaction between hydrogen gas and iodine gas is 138 L mol^{-1} s^{-1}. The activation energy for the reaction is 165 kJ mol^{-1} at 700°C.

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ $\Delta H = -9.48$ kJ mol^{-1}

Sketch and label an energy profile diagram for this reaction.

Determine the activation energy and rate constant for the reverse reaction at 700°C. (Assume Arrhenius factor, *A* is the same for both forward and reverse reactions.) 5 m

Answer (b) The minimum amount of energy required to initiate a chemical reaction



- Axes
- Shape
- Label Ea

E_a (reverse) = 165 – (- 9.48)

= 174.48 kJ/mol

$K = Ae^{-Ea/RT}$ @ $\ln K = \ln A - E_a / RT$

$\ln 138 = \ln A - \dfrac{165 \times 10^3}{8.314 \times 973}$

$A = 9.956 \times 10^{10}$

$\ln K = \ln A - E_a/RT$

$= 25.324 - \dfrac{174.48 \times 10^3}{8.314 \times 973}$

$= 3.755$

$K = 42.75$ L mol^{-1} s^{-1} @ M^{-1} s^{-1}

PSPM 2012 / 2013

Q. 2 (a) ***L*** reacts with ***M*** according to the equation

$$L + M \longrightarrow N$$

The results of the experiment are given in **TABLE 1.**

TABLE 1

Experiment	[***L***]/mol L^{-1}	[***M***]/mol L^{-1}	**Initial rate of reaction**/mol L^{-1} s^{-1}
1	0.001	0.004	0.002
2	0.002	0.004	0.008
3	0.004	0.001	0.008
4	0.004	0.002	0.016

Based on the data given in **TABLE 1**, determine the order of reaction with respect to ***L*** and ***M***, and calculate the rate constant for the reaction.

8 m

Answer (a)

Rate = $k\,[L]^x[M]^y$

$$\frac{rate\ 2}{rate\ 1} = \frac{0.008}{0.002} = \frac{k(0.002)^x(0.004)^y}{k(0.001)^x(0.004)^y} \text{ @}$$

$$\frac{0.008}{0.002} = \left(\frac{0.002}{0.001}\right)^x \text{ @}$$

$4 = (2)^x$

x = 2

The order of reaction with respect to ***L*** = 2

$$\frac{rate\ 3}{rate\ 4} = \frac{0.008}{0.016} = \frac{k(0.004)^x(0.001)^y}{k(0.004)^x(0.002)^y} \text{ @}$$

$$\frac{1}{2} = \left(\frac{1}{2}\right)^y$$

y = 1

The order of reaction with respect to ***M*** = 1

$$k = \frac{rate}{[L]^2\,[M]}$$

$$k = \frac{0.002\ mol\ L^{-1}\ s^{-1}}{(0.001\ mol\ L^{-})^2\,(0.004\ molL^{-1})}$$

$= 5 \times 10^5\ M^{-2}\ s^{-1}$ (unit insist)

Q. 2 (b) Discuss the effect of temperature and activation energy, E_a on the rate constant, k of a reaction using Arrhenius equation.
Explain effect of temperature on reaction rate using Maxwell-Boltzmann distribution curve. 12 m

Answer (b)

$\ln k_1/k_2 = Ea/R\,(1/T_2 - 1/T_1)$

From the equation $k = Ae^{-\frac{Ea}{RT}}$

Increasing T causes **the term** $e^{-\frac{Ea}{RT}}$ **to be larger** ,Thus **increasing *k***

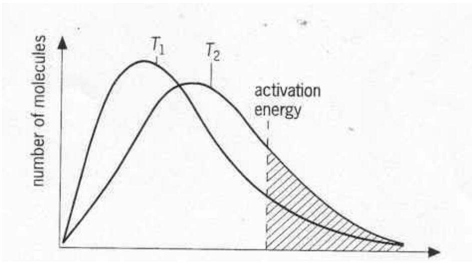
@

Decreasing T causes **the term** $e^{-\frac{Ea}{RT}}$ to **be smaller** thus **decreasing *k*.**

Increasing E_a causes **the term** $e^{-\frac{Ea}{RT}}$ to **be smaller thus decreasing *k*.**

@

Decreasing E_a causes **the term** $e^{-\frac{Ea}{RT}}$ **to be larger thus increasing *k*.**



Molecular speed

When T increases, **effective collision increases**, therefore the **number of molecules having minimum energy to react increase @ total energy ≥ Ea.** Thus the **rate of reaction increase**.
(Maxwell-Boltzmann distribution curve flatten towards higher energy when temperature increase-refer to diagram)

PSPM 2013 / 2014

Q. 3 (a) Define order of reaction and activation energy.

The following reaction is first order with respect to NO_2 and zero order with respect to O_2.

$$2NO_2 + 1/2O_2 \longrightarrow N_2O_5$$

In an experiment, the rate constant, *k*, was determined at different temperatures and the data were tabulated in **TABLE 1**.

TABLE 1

Temperature (K)	k (s^{-1})
477	1.8×10^{-4}
523	2.7×10^{-3}
577	3.0×10^{-2}
667	2.6×10^{-1}

Write the rate law of this reaction and determine graphically the activation energy, E_a, (in kJ mol^{-1}) for the reaction.

In another experiment, the reaction is first order with an activation energy of 111 kJ mol^{-1}. If the rate constant, *k*, is 2.00×10^{-5} s^{-1} at 50°C, determine the frequency factor, A. At what temperature, the value of *k* will be equal to 1.5×10^{-3} s^{-1}?

20 m

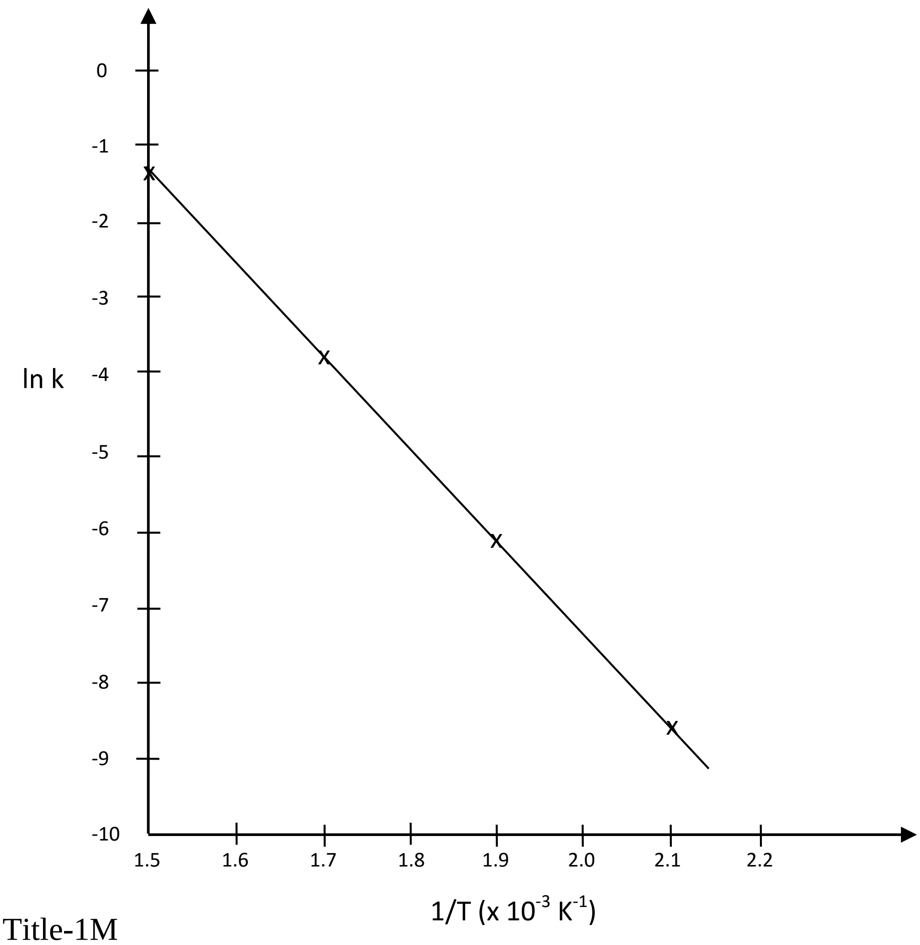
Answer (a)

i) Rate = $k\,[NO_2]^X[O_2]^Y$

Table for the **values 1/T** and **ln *k***

1/T	ln k
2.1×10^{-3}	-8.6
1.9×10^{-3}	-5.9
1.7×10^{-3}	-3.5
1.5×10^{-3}	-1.3

Plot of ln k versus1/T



Title-1M
Shape-1M
y-axis-1M
x-axis-1M
4 correct points-1M

ii) From Arrhenius equation:

$$\ln k = -\frac{E_a}{R}\left(\frac{1}{T}\right) + \ln A$$

$$\therefore slope = -\frac{\mathrm{E_a}}{\mathrm{R}}$$

Slope = -12.167$\times 10^{-3}$ K = $-\frac{E_a}{R}$

$\therefore\ E_a = (12.167 \times 10^{-3})(8.314)$

= 1.01$\times 10^{2}$ $kJ\ mol^{-1}$

iii) $k = Ae^{-\frac{Ea}{RT}}$

$2 \times 10^{-3} = Ae^{-[111000/(8.314 \times 323.15)]}$

$A = 1.754 \times 10^{13}$

OR

$$\ln k = -\frac{E_a}{R}\left(\frac{1}{T}\right) + \ln A$$

$\ln 2 \times 10^{-5} = -\frac{111 \times 10^3 \times 1}{8.324 \times 323.15} + \ln A$

$A = 1.754 \times 10^{13}$

iv) $ln\frac{k1}{k2} = \frac{Ea}{R}(\frac{1}{T2} - \frac{1}{T1})$

$\ln \frac{2 \times 10^{-5}}{1.5 \times x10^{-3}} = \frac{111 \times 10^3}{8.314}(1/T_2 - 1/323.15)$

$T_2 = 360.86$ or $87.71\ ^0C$

PSPM 2014 / 2015

Q. 4 (a) Hydrogen peroxide, H_2O_2, reacts with potassium iodide, KI, in the presence of hydrochloric acid, HCl, according to the following equation:

$$H_2O_2\ (aq)\ +\ 2I^-\ (aq)\ +\ 2H^+ \longrightarrow 2H_2O\ (l)\ +\ I_2\ (s)$$

The rate equation for the reaction is rate = $k\,[H_2O_2][I^-]$.
State the order of reaction with respect to KI and HCl and the overall order of the reaction.

TABLE 2 shows the experimental results obtained for the above reaction. Predict the values of ***G***, ***H*** and ***J*** and calculate the rate constant. If the reaction is carried out at a higher temperature, would the value of the rate constant change?

TABLE 2

Initial rate ($mol\ dm^{-3}\ s^{-1}$)	Initial concentration ($mol\ dm^{-3}$)		
	$[H_2O_2]$	$[H^+]$	$[I^-]$
1.0×10^{-4}	0.1	0.1	0.1
2.0×10^{-4}	***G***	0.1	0.1
1.0×10^{-4}	0.1	0.2	***H***
J	0.2	0.1	0.2

10 m

Answer (a)

KI: First order

HCl: Zero order

Overall: Second order

$\frac{\boldsymbol{rate2}}{\boldsymbol{rate1}} = \frac{[H2O2]^{x}}{[H2O2]^{x}}\frac{[I-]^{y}}{[I-]^{y}}$; $\frac{2.0\ x\ 10^{-4}}{1.0\ x\ 10^{-4}} = \left(\frac{G^1}{0.1^1}\right)\left(\frac{0.1}{0.1}\right)$; ***G*** = 0.2 $mol\ dm^{-3}$,

$\frac{\boldsymbol{rate3}}{\boldsymbol{rate1}} = \frac{[H2O2]^{x}}{[H2O2]^{x}}\frac{[I-]^{y}}{[I-]^{y}}$ $\frac{1.0\ x\ 10^{-4}}{1.0\ x\ 10^{-4}} = \left(\frac{0.1^1}{0.1^1}\right)\left(\frac{H}{0.1}\right)$; ***H*** = 0.1 $mol\ dm^{-3}$,

$\frac{\boldsymbol{rate4}}{\boldsymbol{rate2}} = \frac{[H2O2]^{x}}{[H2O2]^{x}}\frac{[I-]^{y}}{[I-]^{y}}$ $\frac{J}{2.0\ x\ 10^{-4}} = \left(\frac{0.2^1}{0.2^1}\right)\left(\frac{0.2}{0.1}\right)$ ***J*** = $4.0 \times 10^{-4}\ mol\ dm^{-3}\ s^{-1}$

$1.0 \times 10^{-4} = k\,(0.1)\,(0.1)$

$k = 0.01\ dm^3\ mol^{-1}\ s^{-1}$

Yes, the value of *k* will increase.

Q. 4 (b) The rate constant for the reaction below is 0.0234 s^{-1} at 400 °C and 0.750 s^{-1} at 500 °C.
Calculate the activation energy, E_a, in kJ mol^{-1} for this reaction.

$$N_2O\,(g) + NO\,(g) \longrightarrow N_2\,(g) + NO_2\,(g) \qquad \Delta H = -110\ kJmol^{-1}$$

Draw a labeled energy profile diagram for the reaction and calculate E_a for the reversed reaction.

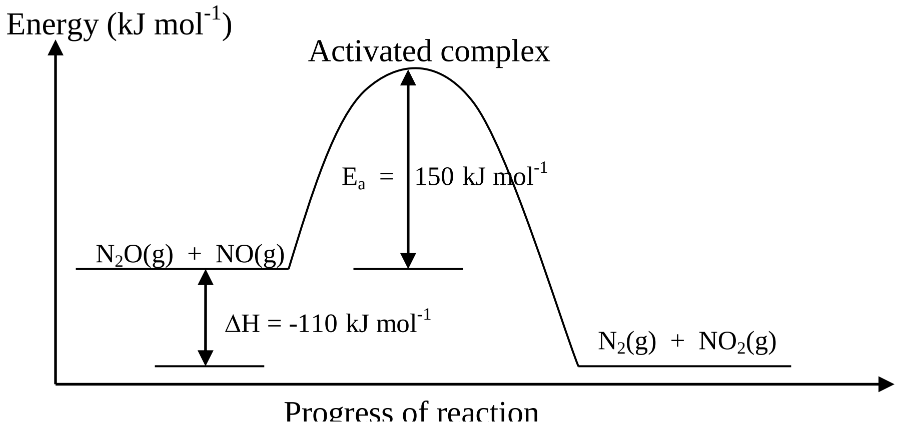
10 m

Answer

$$\ln\frac{k_1}{k_2} = \frac{E_a}{R}\left[\frac{1}{T_2} - \frac{1}{T_1}\right]$$

$$\ln\frac{0.0234}{0.750} = \frac{E_a}{8.314}\left[\frac{1}{773} - \frac{1}{673}\right]$$

E_a = 149 968 J mol^{-1}
= **150 kJ mol^{-1}**



E_a for the reversed reaction = - ΔH (forward) + E_a (forward)
= - (-110 kJ) + 150 kJ
= **+260 kJmol^{-1}**

PSPM 2015 / 2016

Q. 5 (a) The decomposition of hydrogen peroxide, H_2O_2, is according to the following equation: 13 m

$$H_2O_2\,(aq) \longrightarrow H_2O\,(l) + 1/2\,O_2(g)$$

The rate constant for the decomposition of 2.32 mol dm^{-3} H_2O_2 is 7.30 x 10^{-4} s^{-1}.

Plot a graph based on the data in the table below. From the plot, determine the order and half-life of the reaction.

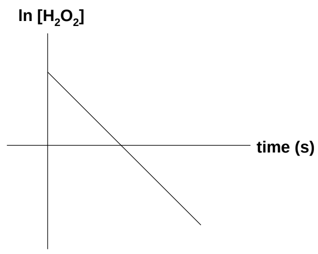
Time (s)	0	200	400	600	1200	1800	3000
$[H_2O_2]$	2.32	2.01	1.72	1.49	0.98	0.62	0.25

Calculate the concentration of H_2O_2 at 1600 s and the percentage of H_2O_2 decomposed up to 1600 s.

Answer

Plot graph ln $[H_2O_2]_t$ versus time

Time/s	0	200	400	600	1200	1800	3000
$[H_2O_2]$	2.32	2.01	1.72	1.49	0.98	0.62	0.25
ln$[H_2O_2]$	**0.84**	**0.70**	**0.54**	**0.40**	**-0.02**	**-0.48**	**-1.39**



Plot ln$[H_2O_2]$ vs time gave a straight line thus the decomposition of H_2O_2 is a 1st order reaction.

Gradient, k = -7.30 x 10^{-4} s^{-1} (from graph)

Half life, $t_{½} = \frac{\ln 2}{k}$

$$= \frac{\ln 2}{7.30 \times 10^{-4}}$$

= 949.5 s

ln $[H_2O_2]_t$ = -kt + ln $[H_2O_2]_0$

ln $[H_2O_2]_0$ – ln $[H_2O_2]_t$ = kt

ln (2.32) – ln $[H_2O_2]_t$ = (7.30 x 10^{-4}) (1600)

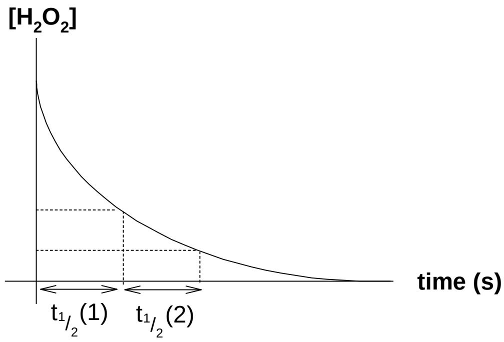
$[H_2O_2]_t$ = **0.72 M**

% decomposition = $\frac{2.32 - 0.72}{2.32}$ x 100%

= **68.9%**

~Alternative answer~

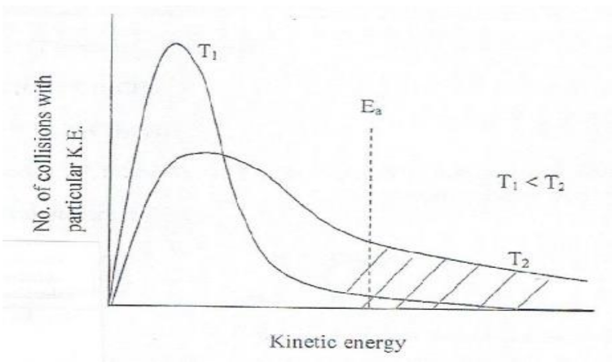
Plot graph $[H_2O_2]$ versus time.



From the graph obtained, 1^{st} $t_{½}$ $\approx$ 2^{nd} $t_{½}$
the decomposition of H_2O_2 is a 1^{st} order reaction.
Therefore, $t_{½}$ = **960 s**.

Q. 5 (b) Both reaction rate and rate constant are temperature dependent. Briefly explain the effect of temperature on reaction rate by using Maxwell-Boltzmann distribution curve. 7m

Answer



As temperature increase,

- More molecules move at higher speed , kinetic energy of molecules increase.
- More molecules having kinetic energy ≥ E_a
- The number of effective collisions increase, thus reaction rate increase.

PSPM 2016 / 2017

Q. 6 (a) Reactant **S** react to form products **T** and **V** at constant temperature according to the following reaction,

$$S \rightarrow T + V$$

In an experiment, the data obtained were shown in **TABLE 1**.

TABLE 1

Time (min)	0	10	20	30	40	50	60
Concentration of **S** (mol dm^{-3})	4.50	3.10	2.38	1.92	1.60	1.40	1.22

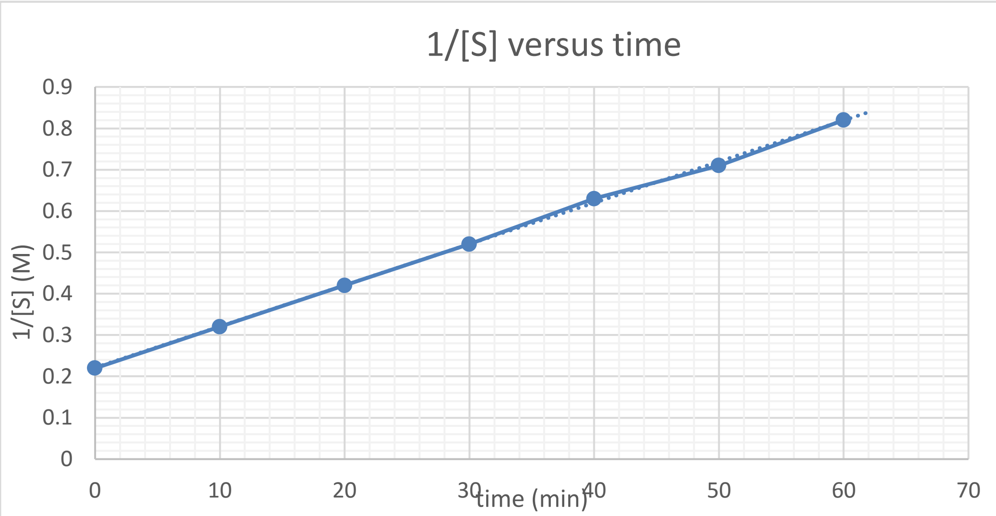
Using integrated rate law method, plot a graph to prove that the above reaction follows second order. Calculate the value of the rate constant, half-life and write the rate law for this reaction. Based on your graph, determine the concentration of S at 1500 s. 10 m

Answer

Integrated rate law for second order reaction is,

$$\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$$

Time (min)	0	10	20	30	40	50	60
[S] (M)	4.50	3.10	2.38	1.92	1.60	1.40	1.22
1/[S]	**0.22**	**0.32**	**0.42**	**0.52**	**0.63**	**0.71**	**0/82**



Label both axes with unit

Straight line with 7 points

A plot of $\frac{1}{[A]_t}$ versus t will give a **straight line**.

☐Therefore, the reaction is proven as 2nd order reaction.

Show horizontal and vertical line for 1500s in the graph.

k = m = $\frac{(0.82-0.22)}{(60-0)}$ = **0.01 M^{-1} min^{-1} @ (0.009 – 0.011) M^{-1} min^{-1}**

@ 1.67 x 10^{-4} $M^{-1}s^{-1}$ @ (1.60 x 10^{-4} – 1.80 x 10^{-4}) $M^{-1}s^{-1}$

$$t_{1/2} = \frac{1}{k[A]_0}$$

$\frac{1}{(0.01)(4.50)}$ = **22.2 min @ 1333.2 s**

Rate = $k[S]^2$ @ rate = $0.01[S]^2$

At 1500 s, $\frac{1}{[S]} = 0.47$

[S] = 2.13 mol dm^{-3}

Q. 6 (b) In a reaction **W** → **X**, the initial rate is 2.25 x 10^{-3}mol $dm^{-3}s^{-1}$ when the concentration of **W** is 0.33 mol dm^{-3}. If the reaction is a first order kinetics, calculate the initial rate of the reaction when the concentration of **W** is 0.85 mol dm^{-3}.

Given the rate constants for the first order reaction at 25°C and 35°C are 1.74 x 10^{-5} s^{-1} and 6.61 x 10^{-5} s^{-1} respectively, determine its activation energy. Other than temperature, discuss **two (2)** factors that would affect reaction rate. 10 m

Answer (b)

For first order reaction:

Rate of reaction α [W] @ rate = k[W]

$$\frac{rate_1}{rate_2} = \frac{(2.25x10^{-3})}{rate_2} = \frac{0.33}{0.85}$$

$rate_2$ = 5.80 x 10^{-3} mol dm^{-3} s^{-1}

$$\ln\left(\frac{k_1}{k_2}\right) = \frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\ln\left(\frac{1.74x10^{-5}}{6.61x10^{-5}}\right) = \frac{E_a}{8.314}\left(\frac{1}{308} - \frac{1}{298}\right)$$

E_a = 101.85 @ 102 kJ mol^{-1}

Concentration

As concentration increase, the frequency of effective collision increase.

@

Pressure

As pressure applied, number of molecules per unit volume increase. The frequency of effective collision increase.

@

Catalyst

Catalyst provide an alternative pathway by lowering the activation energy. Thus, the frequency of effective collision increase.

@

Particle size

The smaller the particle size, the greater the total area exposed for reaction. Therefore effective collision increase.

PSPM 2017 / 2018

Q. 7 N_2O_5 decomposes to NO_2 and O_2 according to the folloeing chemical equation:

$$N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$$

TABLE 1 shows the changes in N_2O_5 concentration against time when the reaction is performed at 50°C.

TABLE 1

$[N_2O_5]$/ M	1.00	0.77	0.60	0.45	0.35	0.27
Time / min	0	1	2	3	4	5

Determine the order and the rate constant for the reaction using a linear graph method. From the graph, what is the half – life of the reaction?

Given the E_a of this reaction is 58 kJ mol^{-1}, calculate the rate constant at 65°C.

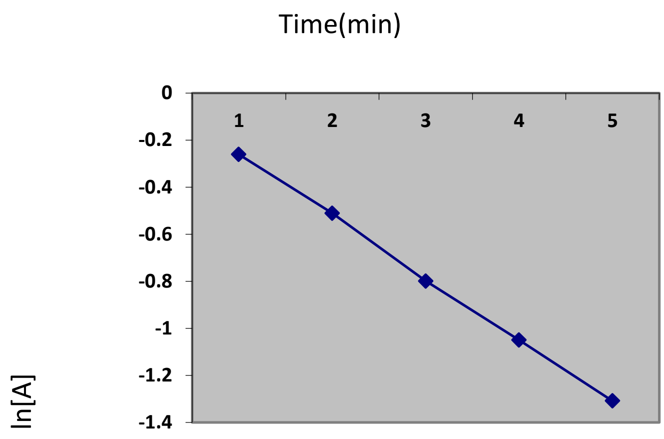
Compare the rate constant at the two temperatures and explain it in terms of collision theory.

Describe how the addition of catalyst affects the rate of the reaction.

20 m

Answer

Time/min	[A]	ln[A]
0	1.00	0
1	0.77	-0.261
2	0.60	-0.511
3	0.45	-0.799
4	0.35	-1.050
5	0.27	-1.309



Straight line, therefore it is 1st order reaction

K = - gradient

= (show from the graph)

= 0.271 min^{-1} 9range from 0.260 to 0.271)

From the graph (show dotted lines --------)

$[A]_{1/2}$ = 0.5

ln 0.5 = -0.693

$t_{1/2}$ = (range 2.56 – 2.67)

@

$t_{1/2} = \frac{\ln 2}{k}$

= 2.56 min (range 2.56 – 2.67)

$$\ln\left(\frac{k_1}{k_2}\right) = \frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\ln\left(\frac{0.271}{k_2}\right) = \frac{5.8\text{x}10^4}{8.314}\left(\frac{1}{338} - \frac{1}{323}\right)$$

k_2 = 0.707 min^{-1} (0.678 – 0.707)

PSPM 2018 / 2019

Q. 8 (a) On heating, dinitrogen pentoxide, N_2O_5 decomposes as follow: 7m

$$2N_2O_5\,(g) \rightarrow 4NO_2\,(g) + O_2\,(g)$$

The reaction is first order with respect to N_2O_5.

(i) Determine the rate of formation of O_2 if the rate of disappearance of N_2O_5 is 1.25×10^{-2} M min^{-1}.

Answer

$$+\frac{d[O_2]}{dt} = -\frac{1}{2}\frac{d[N_2O_5]}{dt}$$

$$= \frac{1}{2} \times \left(1.25 \times 10^{-2}\right)$$

$$= 6.25 \times 10^{-3} \text{ M min}^{-1}$$

(a)(ii) Calculate the rate constant for this reaction if 25% of N_2O_5 is decomposed within 10 minutes.

Answer

Assume $[N_2O_5]_o = 1.0$ M

$$\text{In}\frac{[N_2O_5]_0}{[N_2O_5]_t} = kt$$

$$\text{In}\frac{[1.0]}{[0.75]} = k(10\text{min})$$

$$= 2.87 \times 10^{-2} \text{ min}^{-1} \text{ @ } 4.79 \times 10^{-4} \text{ s}^{-1}$$

(a)(iii) Determine the half-life for this decomposition.

Answer

$$t_{\frac{1}{2}} = \frac{\text{In } 2}{k}$$

$$= \frac{\text{In } 2}{2.87 \times 10^{-2}}$$

= 24.15 min @ 1449 s

(b) Table 1 shows the rate constant for the decomposition of N_2O_5 at two different temperatures. 7m

TABLE 1

Temperature (°C)	Rate constant (s^{-1})
25.0	1.74 x 10^{-5}
35.0	6.61 x 10^{-5}

(i) Determine the activation energy, Ea for this reaction in kJ mol^{-1}.

Answer

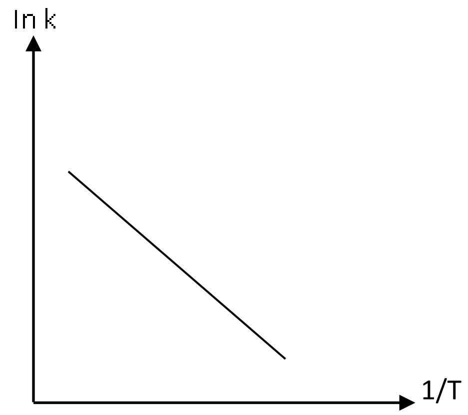
$$\ln\frac{k_1}{k_2} = \frac{E_a}{R}\left[\frac{1}{T_2} - \frac{1}{T_1}\right]$$

$$\ln\left(\frac{1.74\times10^{-5}}{6.61\times10^{-5}}\right) = \frac{Ea}{8.314}\left(\frac{1}{308} - \frac{1}{298}\right)$$

E_a = **101.85 kJ mol^{-1}**

(b)(ii) Activation energy can also be determined graphically. Sketch the graph and explain briefly.

Answer



Negative slope -1M
Axis -1M

From Arrhenius equation:

$$\ln k = -\frac{E_a}{R}\left(\frac{1}{T}\right) + \ln A$$

$$\therefore slope = -\frac{\mathrm{E_a}}{\mathrm{R}}$$

Ea = - slope x R